

combi

Wednesday, 4 June, 2025

8:29 AM

1: WARM-UP

$S \rightarrow n$ elements

number of subsets of S ?

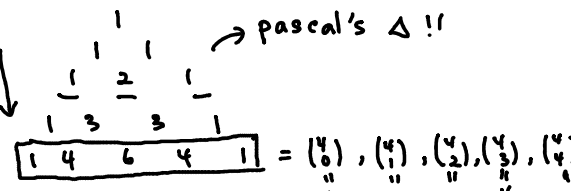
$S = \{1, 2, 3, 4\} \rightarrow 4$ elms
 $\{1, 2, 2, 3\} \times$

$S' \subseteq S$
 $\hookrightarrow$ subset $S' = \{1, 2, 3\}$
 $S' = \{1, 2, 5\} \times$

$S = \{a_1, a_2, \dots, a_n\}$
 n elms $S' = \{\} = \emptyset$
 $S' = \{a_1, a_3\}$
 $S' = \{a_1, a_2, \dots, a_n\}$

$S = \{1, 2, 3\}$
 $\{1, 2\}, \{2, 3\}, \{1, 3\}$
 3 $\{ \}$
 $3C_2 = 3$

no. of subsets of $S = 2^n$
 no. of subsets of S with 2 elms $= \binom{n}{2}$
 $\rightarrow \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$

sum $= 2^n$
 $\rightarrow$ pascal's Δ !!

 $1 \ 4 \ 6 \ 4 \ 1 = \binom{4}{0}, \binom{4}{1}, \binom{4}{2}, \binom{4}{3}, \binom{4}{4}$

COMBINATIONS

$\binom{n}{r} = {}^nC_r \rightarrow$ pick r things from n things

5 ppl

$${}^5C_2 = \frac{5!}{2!3!}$$

5 ppl
pick 3 ppl

$${}^5C_3 = \frac{5!}{3!2!}$$

$$= \frac{120}{6 \times 2}$$

$$= 10$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

4 ppl, pick 2 ppl

$${}^4C_2 = \frac{4!}{2!2!}$$

$$= \frac{24}{2 \times 2}$$

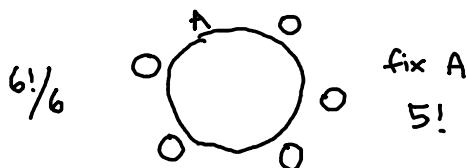
$$= 6$$

How many ways can you
line up 5 ppl in a row?

A B C D E

$$\overline{5 \times 4 \times 3 \times 2 \times 1} = 5! = 120$$

How many ways can you seat
6 ppl around a circular
table, if rotations count as one?



How many ways can you
seat 7 girls and 3 boys
in a row such that no
boy sits next to another boy?

$$10! - \underline{(9! \cdot 2! \cdot {}^3C_2)} - 8! \cdot 3! = 1209600$$

↳ 2 boys next to each other

↳ doesn't stop
all 3 boys together

count directly

_ G _ G _ G _ G _ G _ G _

arrange girls $\rightarrow 7!$

insert boys $\rightarrow {}^8C_3$

arrange boys $\rightarrow 3!$

$$\text{final ans} = 7! \times {}^8C_3 \times 3!$$

$$= 1693440 \checkmark$$

How many 4-digit positive integers contain at least one even digit?

* complementary counting

$$9000 - (5 \times 5 \times 5 \times 5) = 8375$$

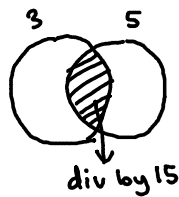
1, 3, 5, 7, 9

How many positive integers less than 100 are divisible by 3 or 5?

$$\text{div by 3} \rightarrow 33$$

$$\text{div by 5} \rightarrow 19$$

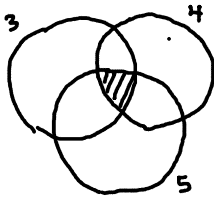
$$\text{div by 15} \rightarrow 6$$



$$33 + 19 - 6 = 46$$

* principle of inclusion and exclusion

How many positive integers less than 300 are divisible by 3, 4 or 5?



$$\text{div by 3} \rightarrow 99$$

$$\text{div by 4} \rightarrow 74$$

$$\text{div by 5} \rightarrow 59$$

$$\text{div by 12} \rightarrow 24$$

$$\text{div by 15} \rightarrow 19$$

$$\text{div by 20} \rightarrow 14$$

$$\text{div by 60} \rightarrow 4$$

$$3 \times 4 \times 5$$

$$! 60$$

final ans

$$= (99 + 74 + 59)$$

$$- (24 + 19 + 14) + 4$$

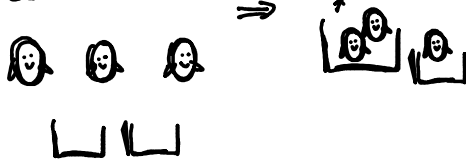
$$= 179$$

2: PIGEONHOLE PRINCIPLE (PHP)

$n+1$ pigeons
 n holes $\Rightarrow \exists$ hole that has > 1 pigeon

$n+1$ pigeons
 n holes $\Rightarrow \exists$ hole
 that has
 >1 pigeon

3 pigeons
 2 holes $\Rightarrow$ >1 Pigeon



generalised:

m pigeons
 n holes $\Rightarrow \exists$ hole that
 has $\geq \lceil \frac{m}{n} \rceil$ pigeons

Prove that among 400 ppl,
 there must be two of them
 who share the same birthday.

400 birthdays $\rightarrow$ pigeons

366 dates $\rightarrow$ holes

$400 > 366$ true by PHP

inf many
 blue socks
 red socks
 yellow socks

minimum no.
 of socks to pull
 out to guarantee
 a pair

4 socks $\rightarrow$ pigeons

3 colours $\rightarrow$ holes

Prove that in a group
 of people, there are two who
 have the same number of
 friends within that group.

n ppl

each person has $0, 1, 2, \dots$ or $n-1$
 friends

Case 1: no one has 0 friends

each person has 1, 2, ... or $n-1$
friends

$n-1$ possibilities $\rightarrow$ holes

n ppl

$a_1, a_2, a_3, \dots, a_n \rightarrow$ pigeons

by PHP, there are two ppl
w/ the same number of friends

Case 2: there is a person with 0 friends

each person has 0, 1, 2, ... or $n-2$
friends ~~$n-1$~~

$n-1$ possibilities $\rightarrow$ holes

n ppl

$a_1, a_2, \dots, a_n \rightarrow$ pigeons

by PHP, there are two ppl
w/ the same number of friends

Suppose S is a set of $n+1$ integers.

Prove there exists distinct $a, b \in S$ NT
s.t. $n \mid a-b$ flavour

$a_1, a_2, a_3, \dots, a_{n+1} \rightarrow$ pigeons

$\text{mod } n \rightarrow 0, 1, 2, \dots, n-1 \rightarrow$ holes
 $\rightarrow n$ possibilities

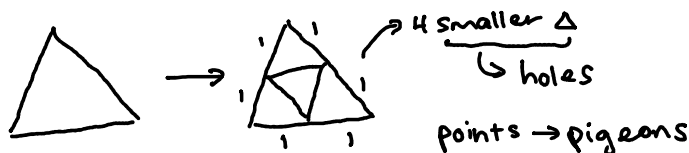
$a_i \equiv a_j \pmod{n} \Rightarrow a_i - a_j \equiv 0 \pmod{n}$
by PHP $\Rightarrow n \mid a_i - a_j$

G flavour

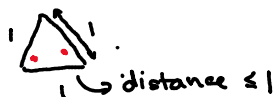
5 points are placed inside an equilateral triangle with side length 2. Prove that there is a pair of points such that the distance between them is at most 1.



hint:
divide
the Δ



by PHP, one of the small Δ s contains ≥ 2 points



Show that for any $n \in \mathbb{N}$, there exists a positive multiple of n that only contains the digits 1 and 0. (with at most n digits)

hint: focus on numbers only containing the digit 1

1, 11, 111, 1111, 11111, ..., $\underbrace{111 \dots 11}_{n \text{ digits}}$

mod n ?

case 1: there is a number $0 \pmod n$ done

case 2: there is no number $0 \pmod n$

1, 2, ..., $n-1 \rightarrow n-1$ possibilities (holes)

n numbers (pigeons)

by PHP, $a \equiv b \pmod n$

$a-b$ mult. of n

$$11111 - 111 = 111000$$

Prove there exists a subset of n integers (not necessarily distinct) whose sum is divisible by n .

$$a_1, a_2, a_3, \dots, a_n$$

$$a_1, a_1+a_2, a_1+a_2+a_3, \dots, a_1+a_2+\dots+a_n$$

n numbers

mod n

case 1: there is a number $0 \pmod n$
done

case 2: there is no number $0 \pmod n$

$$1, 2, \dots, n-1 \rightarrow n-1 \text{ possibilities (holes)}$$

n numbers (pigeons)

by PHP, $a \equiv b \pmod n$

$a-b$ mult. of n

$$a_1 + a_2 + a_3 + a_4 \equiv a_1 + a_2 \pmod n$$

$$\Rightarrow \underbrace{a_3 + a_4}_{\text{still a sum!!}} \equiv 0 \pmod n$$

JMO SL 2013 C1

find the maximum number of different integers that can be selected from the set $\{1, 2, \dots, 2013\}$ so that no two selected integers differ by 17.

hint: 2006 is a multiple of 17

$$0 \pmod{17} \rightarrow 17, 34, \dots, 2006 \rightarrow 118 \text{ numbers}$$

$$= 17 \times 1, 17 \times 2, 17 \times 3, 17 \times 4, \dots, 17 \times 117, 17 \times 118$$

max 59 $\rightarrow$ alternating

$$\begin{array}{l} 1 \pmod{17} \rightarrow 1, 18, 35, \dots, 2007 \\ 2 \pmod{17} \rightarrow 2, 19, 36, \dots, 2008 \\ \vdots \\ 16 \pmod{17} \rightarrow 16, 33, \dots, 2005 \end{array} \left. \begin{array}{l} \rightarrow \text{max } 60 \\ 1-7 \pmod{17} \\ 119 \text{ numbers} \\ 8-16 \pmod{17} \\ 118 \text{ numbers} \\ \rightarrow \text{max } 59 \end{array} \right\}$$

$$60(7) + 59(10) = 1010$$

Manhattan MO 2005

Prove that having 100 integers, we

Manhattan NK - - - -

Prove that having 100 integers, we can choose 15 of them so that the difference of any two is divisible by 7.

hint: mod 7

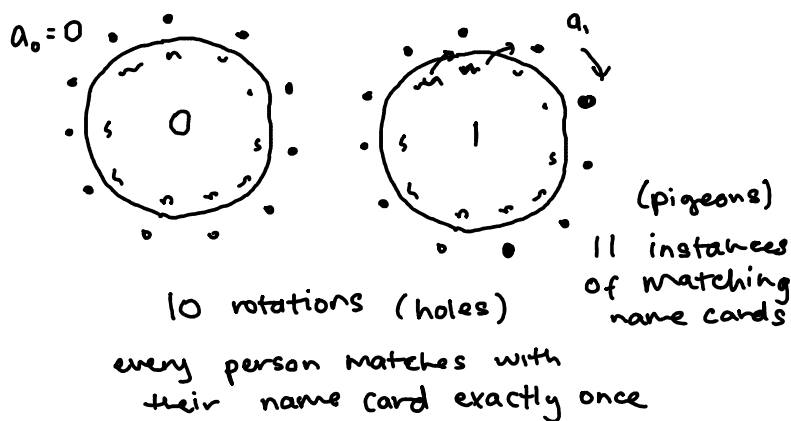
0 mod 7
 1 mod 7
 2 mod 7
 3
 4
 5
 6
 holes

100 integers
 pigeons

$\exists$ hole with $\geq \left\lceil \frac{100}{7} \right\rceil$ pigeons
 15

3. (JBMO SL 2012 C1) Along a round table are arranged 11 cards with the names (all distinct) of the 11 members of the 16th JBMO Problem Selection Committee. The cards are arranged in a regular polygon manner. Assume that in the first meeting of the Committee none of its 11 members sits in front of the card with his name. Is it possible to rotate the table by some angle so that at the end at least two members sit in front of the card with their names?

hint: think abt all 10 possible rotations



$$a_1 + a_2 + \dots + a_{10} = 11$$

YES! $\exists$ a rotation where ≥ 2 name cards match $\therefore$

4. (JBMO SL 2008 C3) Integers $1, 2, \dots, 2n$ are arbitrarily assigned to boxes labeled with numbers $1, 2, \dots, 2n$. Now, we add the number assigned to the box to the number on the box label. Show that two such sums give the same remainder modulo $2n$.

hint: assume all sums give different

sums give the same remainder modulo $2n$.

hint: assume all sums give different remainders mod $2n$

$$\hookrightarrow 0, 1, 2, \dots, 2n-2, 2n-1 \\ \text{mod } 2n$$

$$0+1+2+\dots+2n-1 \equiv \frac{2n(2n-1)}{2} \equiv n(2n-1) \equiv n(-1) \pmod{2n} \\ \equiv -n \equiv \underline{n}$$

$$2[1+2+\dots+2n] = 2\left(\frac{2n(2n+1)}{2}\right) = \underline{2n(2n+1)} \equiv \underline{0} \pmod{2n}$$

contradiction

Let $S = \{1, 2, \dots, 2n\}$.

Choose $n+1$ numbers from S .

Prove that one number will be the multiple of another

$$\begin{array}{l} \rightarrow 2^a b \\ \quad \downarrow \\ \quad \text{odd} \\ \quad \downarrow \\ n \text{ possibilities} \\ \quad \text{(holes)} \end{array} \quad \begin{array}{lll} 1 = 2^0 & 4 = 2^2 & 7 = 2^0 7 \\ 2 = 2^1 & 5 = 2^0 5 & 8 = 2^3 \\ 3 = 2^0 3 & 6 = 2^1 3 & 9 = 2^0 3^2 \end{array}$$

$n+1$ numbers
(pigeons)

by PHP, $\exists$ two numbers with the same b

$\hookrightarrow$ one is a multiple of the other